
Affine-Invariant SPD Action Completion: Global Geometry, Posterior Certificates, and Active Complexity

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Abstract

We study affine-invariant least-deformation completion of a positive-definite operator from its prescribed action on a subspace. For full-column-rank matrices $A, B \in \mathbb{R}^{d \times m}$, the principal problem minimizes the squared affine-invariant Riemannian distance from a reference over determinant-one $P \succ 0$ satisfying $PA = B$. We first formulate hidden-state elimination by infimal pushforward. It composes across nested maps, and in fixed product coordinates its smooth sensitivity is governed by a Schur complement; affine perturbations before elimination induce the negative-semidefinite interaction curvature $-G^*H^{-1}G$. For the action map $P \mapsto PA$, we prove a global analytic bundle whose fibers are closed and totally geodesic, yielding a unique analytic nearest-controller section. An explicit logarithmic residual then generates a globally linearly convergent iteration from every initial hidden coordinate, with nonasymptotic value, distance, and residual bounds and computable a posteriori stopping certificates. The guarantees extend conditionally when a residual oracle supplies certified direction-error bounds. From the identity, or any hidden state compatible with the active symmetry, the iteration has an exact realization on a subspace of dimension $r \leq 2m$, reducing its dense spectral kernel to $r \times r$ operations. Controlled synthetic experiments verify the certificates, full-active equivalence, dimension reduction, and deterioration of the radius constant near the feasibility boundary. Finally, nested shape-normalized multi-secant projections satisfy a CAT(0) Pythagorean decrease; a residual-controlled version covers finite inner solves, and the fibers recover the determinant-one inverse-Hessian shape at the sharp cumulative rank $d - 1$, given its scalar determinant gauge.

1 Introduction

Let $A, B \in \mathbb{R}^{d \times m}$, with A of full column rank and $1 \leq m < d$. Consider the affine-invariant least-deformation problem

$$\min_{P \in \mathbb{S}_{++}^d} \frac{1}{2} d_{\text{AI}}(P_{\text{ref}}, P)^2 \quad \text{subject to} \quad PA = B, \quad \det P = \det P_{\text{ref}}. \quad (1)$$

After an affine-invariant congruence and a scalar normalization, it is enough to take $P_{\text{ref}} = I$ and $\det P = 1$. The feasibility condition is $A^\top B \in \mathbb{S}_{++}^m$. When $m < d$, the action $PA = B$ leaves a positive-dimensional family of feasible completions, and equation (1) selects the one of minimum intrinsic deformation. The same problem is an affine-invariant multi-secant completion when A contains secant directions and B contains their desired inverse actions.

This completion problem is motivated by adaptive optimization. Moments, curvature surrogates, Kronecker factors, and tensor preconditioners act as hidden positive operators that turn visible gradients into parameter motion Duchi et al. (2011); Kingma & Ba (2015); Martens & Grosse (2015); Gupta et al. (2018); Morwani et al. (2024); Jordan (2024); Essential AI et al. (2025); Crawshaw et al. (2025). Observing the action of such an operator on a low-dimensional subspace does not determine its complementary geometry. Problem

equation 1 supplies a canonical completion and an intrinsic measure of the deformation required to realize the observed action.

Three issues prevent the formulation alone from giving an algorithmic result. The feasible set is a nonlinear SPD fiber whose global topology and geodesic geometry are initially unknown; nearest-point existence does not by itself produce a computable optimality residual; and a dense implementation would appear to require matrix functions in the ambient dimension d . We resolve these issues by combining a global action-bundle theorem with an explicit residual-gradient iteration and an exact active-subspace reduction.

The matrix problem sits inside a general variational-reduction mechanism. For hidden state x , visible action $y = \pi(x)$, and deformation energy E , define the infimal pushforward

$$(\pi_! E)(y) = \inf_{\pi(x)=y} E(x).$$

It eliminates hidden realizations and composes across nested maps. Convex-fiber regularity selects a canonical lift, while smooth nondegeneracy gives

$$D\sigma_* = -E_{\sigma\sigma}^{-1}E_{\sigma y}, \quad D^2(\pi_! E) = E_{yy} - E_{y\sigma}E_{\sigma\sigma}^{-1}E_{\sigma y}.$$

When mechanism amplitudes enter affinely before reduction, their interaction curvature is

$$D_{uu}^2(\pi_! E) = -G^* H^{-1} G \preceq 0,$$

and its mixed entries integrate to finite contrasts under the specified intervention protocol. In the SPD action problem, the same vertical Hessian governs both this response and the susceptibility of the nearest completion.

Our principal realization is the determinant-one SPD action map

$$\pi_A(P) = PA.$$

Every admissible fiber has a global analytic parameterization by a smaller determinant-one SPD factor and is closed and totally geodesic under the affine-invariant metric. Projection of the reference therefore gives a unique analytic canonical controller. A strongly-convex fiber argument connects this selection to computation, while the action bundle makes its logarithmic residual explicit. A sublevel first-return argument yields the radius majorant $L_0 = \psi(D_0/\sqrt{2})$, nonasymptotic iteration bounds from exact inversion of the linear rate, and posterior controller and objective certificates. The same analysis propagates certified matrix-function errors and radius majorants.

From the identity hidden coordinate, and more generally from an active-compatible hidden coordinate, the solver has an exact realization on $\text{span}(\text{col } A, \text{col } B)$, of dimension $r \leq 2m$. It replaces the full spectral kernel by $r \times r$ operations and gives a strict dimension reduction when $r < d$. Arbitrary hidden initialization retains the global convergence theorem but need not preserve this active symmetry. Nested shape-normalized actions then satisfy a CAT(0) Pythagorean law and identify $c_H H^{-1}$ at the sharp rank $d - 1$, provided the scalar gauge $c_H = (\det H)^{1/d}$ is known. Finite inner solves obey an observable residual-controlled perturbation of the same law.

Contributions.

1. **Global geometry of the action constraint.** We construct a global analytic trivialization of $P \mapsto PA$, prove that every determinant-one feasible fiber is closed and totally geodesic, and obtain a unique analytic affine-invariant nearest-completion section.
2. **A global solver with posterior certificates.** The exact logarithmic residual is the whitened negative gradient of a strongly geodesically convex reduced objective. Its exponential update converges globally and linearly from every initial hidden coordinate, with explicit iteration bounds and sharp posterior distance and objective-gap certificates. A conditional inexact theorem propagates certified errors in the residual direction; update, factorization, and storage errors remain the responsibility of the numerical implementation.

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3. **Active complexity and finite identification.** From an active-compatible initializer, the exact iteration reduces to an active space of dimension $r \leq 2m$, costs $O(r^3)$ per step after $O(dm^2 + r^3)$ preprocessing, and applies the resulting controller in $O(dr + r^2)$. Nested projections obey exact and residual-controlled approximate CAT(0) laws and recover determinant-one inverse-Hessian shape after $\lceil (d-1)/b \rceil$ independent width- b rounds, assuming the scalar determinant gauge is supplied.
 4. **Variational response and interaction.** Associative hidden-state elimination yields the canonical response and the Schur reduced Hessian in fixed product coordinates. Affine mechanism perturbations give a negative-semidefinite interaction curvature whose action specialization is locally controlled by the same inverse vertical Hessian as completion sensitivity.

Controlled synthetic experiments accompany the theorem chain. They test the posterior inequalities against high-accuracy reference solutions, compare the full and active iterates, measure the dense spectral kernels as d grows, and track the curvature radius as $\lambda_{\min}(A^\top B) \downarrow 0$.

Together, these results turn the intrinsic completion problem equation 1 into a globally solvable reduced optimization problem with observable termination criteria. The global action bundle links its geometry, sensitivity, algorithm, and finite-identification law through one variational-reduction chain.

Organization. Sections 3 and 4 develop variational reduction and its response theory. Section 5 establishes the global action geometry, section 6 gives the solver, certificates, and operation bounds, section 7 reports controlled numerical tests, and section 8 proves finite recovery. The appendices contain complete proofs and a theorem-level comparison with the closest literature.

2 Related Work

Adaptive geometry and variational reduction. Natural gradient, K-FAC, AdaGrad, Adam, Shampoo, SOAP, and related methods use Fisher, moment, Kronecker, or tensor states as changing positive update operators Amari (1998); Martens & Grosse (2015); Duchi et al. (2011); Kingma & Ba (2015); Gupta et al. (2018); Morwani et al. (2024); Vyas et al. (2024). Riemannian optimization provides the coordinate-free gradient language Absil et al. (2008); Boumal (2023). Partial minimization, marginal functions, envelope sensitivity, and variable projection are classical Rockafellar (1970); Rockafellar & Wets (1998); Bonnans & Shapiro (2000); Golub & Pereyra (1973). In fixed affine parameter coordinates, the Schur complement is the reduced coordinate Hessian of a marginal objective. Global first-order complexity for geodesically convex objectives on Hadamard manifolds is established in Zhang & Sra (2016), and metric projection and convex optimization in Hadamard spaces are developed systematically by Bačak (2014). Our use of these tools is tied to the action constraint: the global fiber coordinates expose an exact logarithmic residual, a current-sublevel curvature majorant attained by the action family, an active realization, and posterior controller certificates.

AIRM approximation and totally geodesic submanifolds. The affine-invariant SPD geometry is classical Bhatia (2007); Pennec et al. (2006). AIRM approximation from special sets, best approximation in geodesic SPD submanifolds, and general total-geodesy criteria and projections are studied by Bhatia & Jain (2014), Lim (2004), and Tumpach & Larotonda (2024); conic geodesic optimization supplies a broader algorithmic SPD setting Sra & Hosseini (2015). Algebraic identities and definiteness conditions for Hermitian solutions of $AX = B$ are studied by Tian (2013). These results supply feasibility and block-Schur algebra, while Hadamard projection theory supplies uniqueness once a fiber is known to be closed and geodesically convex. Lim (2004) additionally gives explicit formulas for selected involutive block models, including low-dimensional block cases.

The comparison is result-specific. Tian’s matrix-equation analysis supplies algebraic existence, definiteness, and solution identities for Hermitian equations; our $A^\top B \succ 0$ condition is used at that feasibility level, whereas the determinant-one AIRM minimizer and its iteration are variational objects. Lim studies best approximation in Riemannian geodesic SPD submanifolds and gives explicit formulas for selected involutive block models; we use the same nearest-point geometry only after proving that every action-indexed fiber is closed and totally geodesic. Tumpach and Larotonda give general total-geodesy criteria and structural results

for SPD submanifolds; our contribution identifies the full $PA = B$ family, constructs its global analytic bundle over B , and derives the normal equation. None of these three results supplies the action residual iteration, its current-sublevel rate and posterior certificates, the active-compatible $r \leq 2m$ realization, or the rank- $d - 1$ normalized recovery law. Conversely, projection uniqueness and the general variational/Schur principles are used as established ingredients rather than claimed as new. The theorem-level map in section D records these boundaries.

Secant updates and positive-definite completion. Inverse-Hessian multi-secant equations have the form $PY = S$ Schnabel (1983); compatibility with positive definiteness and modern block or grouped updates have been studied separately Passy & Prisman (1984); Gao & Goldfarb (2018); Boutet et al. (2022). Classical least-change and variational BFGS/DFP updates use weighted norms or log-determinant divergences Dennis & Schnabel (1979); Fletcher (1991); Bregman and weighted secant extensions use related selection principles Kanamori & Ohara (2013); Gratton et al. (2015). Bounded and sparse quasi-Newton updates and positive-definite completion are treated in Calamai & Moré (1987); Yamashita (2007); Dai & Yamashita (2014); Grone et al. (1984).

Our action formulation gives a global analytic trivialization of $P \mapsto PA$, identifies its determinant-one fibers, and proves their total geodesy before applying the nearest-point principle. It then yields a globally convergent residual solver and the sharp completion rank for shape-normalized nested actions. The rank- $d - 1$ theorem uses a supplied determinant scale, making its information model different from ordinary scale-free secant observations. Condition-constrained SPD approximation and preconditioning have separate literatures Tanaka & Nakata (2014); Maréchal & Ye (2009); Lu & Pong (2011); Gao et al. (2026); Qu et al. (2025); Doğan et al. (2025). A theorem-level comparison is given in section D.

Scope. The companion information-compression map $P \mapsto A^\top PA$ has different fibers (Li, 2026). The present paper is self-contained and treats the full-SPD exact-action problem. Structured controller families, noisy secants, unknown scalar gauges, historical state estimation, and regret lead to distinct constrained-optimization and statistical problems.

3 Variational Reduction of Hidden Geometry

This section isolates the operation that turns hidden optimizer geometry into a visible action theory. No smoothness is needed for the algebraic layer; geometry enters when one asks for a canonical representative of each fiber.

3.1 Infimal pushforward

Let $\pi : \mathcal{X} \rightarrow \mathcal{Y}$ be an arbitrary observation map and let $E : \mathcal{X} \rightarrow (-\infty, +\infty]$ be an extended energy that never takes $-\infty$. We use $\inf \emptyset = +\infty$.

Definition 1 (Infimal pushforward). The infimal pushforward of E along π is

$$(\pi_! E)(y) := \inf_{\pi(x)=y} E(x). \quad (2)$$

Its effective domain is the set of visible actions that admit a finite-energy hidden realization.

Theorem 1 (Exact reduction and composition). Let $\rho : \mathcal{Y} \rightarrow \mathcal{Z}$ be another map. Then

$$(\rho \circ \pi)_! E = \rho_! (\pi_! E). \quad (3)$$

For every $\ell : \mathcal{Y} \rightarrow (-\infty, +\infty]$, using the extended-value convention $+\infty + a = +\infty$ (neither summand takes $-\infty$),

$$\inf_{x \in \mathcal{X}} \{E(x) + \ell(\pi(x))\} = \inf_{y \in \mathcal{Y}} \{(\pi_! E)(y) + \ell(y)\}. \quad (4)$$

Moreover, for a visible potential $\varphi : \mathcal{Y} \rightarrow (-\infty, +\infty]$ under the same convention and a constraint $\mathcal{C} \subseteq \mathcal{X}$,

$$\pi_!(E + \varphi \circ \pi) = \pi_!E + \varphi, \quad (5)$$

$$\pi_!(E + \iota_{\mathcal{C}})(y) = \inf_{\substack{\pi(x)=y \\ x \in \mathcal{C}}} E(x). \quad (6)$$

The theorem is a typed regrouping law. Its value is compositional: a hierarchical optimizer may eliminate full geometry, structured factors, visible action scale, and normalized direction in stages without changing the final reduced energy. It does not by itself guarantee a minimizer, regularity, or a useful geometry on the visible space.

3.2 Canonical minimizer sections

We next give sufficient conditions for the reduced value to select one hidden state. Let $(\mathcal{X}, d)$ be a proper Hadamard space and write $\mathcal{F}_y = \pi^{-1}(y)$.

Assumption 1 (Convex-fiber regularity). Every nonempty fiber $\mathcal{F}_y$ is closed and geodesically convex. The energy E is lower semicontinuous, and its restriction to each fiber is coercive—its fiber sublevel sets are bounded in the induced metric—and μ -strongly geodesically convex for a common $\mu > 0$.

Theorem 2 (Canonical hidden geometry). *Under Assumption 1, every y with finite $(\pi_!E)(y)$ has a unique minimizer*

$$s_E(y) = \arg \min_{\pi(x)=y} E(x), \quad E(s_E(y)) = (\pi_!E)(y). \quad (7)$$

If a group G acts isometrically on $\mathcal{X}$ and acts on $\mathcal{Y}$, if $E(gx) = E(x)$, and if $\pi(gx) = g\pi(x)$ for every $g \in G$, then

$$s_E(gy) = gs_E(y). \quad (8)$$

The minimizer section is energy-dependent. It should be distinguished from the shortest section of a metric submetry, which is selected by distance alone. The two coincide when $E(x) = \frac{1}{2}d(x_0, x)^2$ and the relevant shortest completion is unique.

4 Hidden Response and Interaction Curvature

The algebraic reduction becomes a differential theory when the canonical minimizer varies smoothly. We fix a local product chart throughout this section, and every ordinary second derivative below is taken in that chart. The Schur formula is therefore a coordinate Hessian identity. It defines an intrinsic bilinear form at a full reduced critical point $D_z \bar{E} = 0$, or after ordinary derivatives are replaced by covariant derivatives for specified connections. Fiber stationarity $E_{\sigma} = 0$ alone does not make the visible coordinate Hessian a tensor. The mechanism amplitudes below carry a fixed affine structure, under which their Hessian transforms by congruence.

4.1 Smooth response and effective curvature

Let $z = (y, u)$ range over a finite-dimensional manifold locally represented in Euclidean coordinates, and let $\mathcal{S}$ be a finite-dimensional smooth manifold with local hidden coordinate σ . Consider

$$E : (z, \sigma) \mapsto E(z, \sigma) \in \mathbb{R}.$$

Assumption 2 (Smooth stable reduction). For some $r \geq 2$, the energy is C^{r+1} . It is locally uniformly inf-compact in σ : for every compact parameter set K and every $c \in \mathbb{R}$, the union of the fiber sublevel sets

$$\{\sigma : \exists z \in K, E(z, \sigma) \leq c\}$$

is relatively compact. Every z has a unique interior minimizer $\sigma_{\star}(z)$, and

$$H(z) := E_{\sigma\sigma}(z, \sigma_{\star}(z))$$

is positive definite.

Theorem 3 (Hidden response and Schur reduced Hessian). *Under Assumption 2, the minimizer section and reduced energy*

$$\sigma_*(z) = \arg \min_{\sigma} E(z, \sigma), \quad \bar{E}(z) = E(z, \sigma_*(z))$$

are C^r . At the canonical lift,

$$D_z \sigma_* = -H^{-1} E_{\sigma z}, \quad (9)$$

$$D_z \bar{E} = E_z, \quad (10)$$

$$D_{zz}^2 \bar{E} = E_{zz} - E_{z\sigma} H^{-1} E_{\sigma z}. \quad (11)$$

If $\sigma = (\alpha, \beta)$, let $\mathbb{H}$ denote the full Hessian in (z, α, β) at the canonical lift. When its joint hidden block is positive definite, sequential and joint second-order elimination agree:

$$\text{Schur}_{(\alpha, \beta)} \mathbb{H} = \text{Schur}_{\alpha} (\text{Schur}_{\beta} \mathbb{H}). \quad (12)$$

The response formula is familiar from parametric optimization and variable projection Golub & Pereyra (1973); Bonnans & Shapiro (2000). Here it supplies the local law inside a globally defined geometric action bundle and composes with the optimizer hierarchy in Theorem 1.

4.2 Affine perturbations before reduction

Let $u \in \mathcal{U} \subset \mathbb{R}^p$ denote continuous mechanism amplitudes. Suppose

$$E(y, \sigma, u) = E_0(y, \sigma) + \sum_{i=1}^p u_i \Delta_i(y, \sigma). \quad (13)$$

At the canonical lift define

$$G : T_u \mathcal{U} \rightarrow T_{\sigma}^* \mathcal{S}, \quad Ga = \sum_{i=1}^p a_i D_{\sigma} \Delta_i. \quad (14)$$

Theorem 4 (Reduction-induced interaction curvature). *Under Assumption 2 and equation (13),*

$$\boxed{D_{uu}^2 \bar{E} = -G^* H^{-1} G \preceq 0.} \quad (15)$$

Equivalently,

$$\partial_{u_i u_j}^2 \bar{E} = -\langle D_{\sigma} \Delta_i, H^{-1} D_{\sigma} \Delta_j \rangle. \quad (16)$$

Thus the self-curvature of every mechanism is nonpositive. A pointwise mixed interaction vanishes exactly when the corresponding vertical perturbations are orthogonal in the inverse-Hessian metric. Mixed entries may have either sign.

Even without smoothness, $u \mapsto \bar{E}(y, u)$ is concave wherever finite, because it is a pointwise infimum of affine functions. Smoothness identifies the exact curvature responsible for that concavity.

The affine hypothesis is essential for the sign. With general smooth u -dependence, equation (11) gives

$$D_{uu}^2 \bar{E} = E_{uu} - E_{u\sigma} H^{-1} E_{\sigma u}, \quad (17)$$

so the direct curvature E_{uu} need not be negative semidefinite. The quadratic form is covariant under an affine reparameterization $u = T\tilde{u} + u_0$, transforming as $D_{\tilde{u}\tilde{u}}^2 \bar{E} = T^* D_{uu}^2 \bar{E} T$. Individual mixed entries therefore have meaning only after the mechanism basis, direction, and units are fixed; all comparisons in this paper refer to the specified amplitude protocol.

Factorial finite differences are iterated integrals of the corresponding mixed derivatives; Corollary 4 states the general identity. In particular,

$$\delta_{\{i, j\}} \bar{E} = - \int_0^1 \int_0^1 \langle D_{\sigma} \Delta_i, H^{-1} D_{\sigma} \Delta_j \rangle du_i du_j, \quad (18)$$

where all quantities are evaluated along the canonical minimizer with the other amplitudes fixed at zero. The integral may vanish by cancellation even when pointwise interaction does not. Reduction must precede finite differencing; the counterexample and the exact commuting special case are given in section A.5.

4.3 Action-fiber specialization

The global action chart makes the same response tensor computational. For fixed $B \in \mathcal{B}_A$, perturb the canonical fiber objective by

$$F_B(\Sigma, u) = f_B(\Sigma) + \sum_{i=1}^p u_i \Delta_i(B, \Sigma), \quad \Sigma \in \mathcal{P}_{d-m}. \quad (19)$$

Let $H_B = \text{Hess } f_B(\Sigma_*)$ and $G_B a = \sum_i a_i D_\Sigma \Delta_i(B, \Sigma_*)$. Strong convexity of the action objective gives $H_B \succeq \text{Id}$ as a self-adjoint operator with respect to the AIRM metric. For an intrinsic perturbation, we require the natural complement-frame covariance $\Delta_{i,VR}(B, R^\top \Sigma R) = \Delta_{i,V}(B, \Sigma)$; otherwise the chosen Δ_i deliberately selects a hidden frame. Derivatives with respect to B use the natural affine structure of the open matrix domain $\mathcal{B}_A \subset \mathbb{R}^{d \times m}$.

Corollary 1 (Interaction response of the canonical action controller). *Assume the Δ_i are C^2 near Σ_* . For sufficiently small u , there is a unique critical point $\Sigma_*(B, u)$ near Σ_* ; it is a strict local minimizer. Define the corresponding local value branch by $\bar{F}_B^{\text{loc}}(u) = F_B(\Sigma_*(B, u), u)$. Then*

$$D_u \Sigma_*(B, 0) = -H_B^{-1} G_B, \quad D_{uu}^2 \bar{F}_B^{\text{loc}}(0) = -G_B^* H_B^{-1} G_B. \quad (20)$$

The same inverse vertical Hessian H_B^{-1} governs the hidden-coordinate response to the visible action B . The full section derivative also contains the chart's direct term:

$$D_B s_A = D_B \Phi_A + D_\Sigma \Phi_A [D_B \Sigma_*].$$

Thus interaction curvature, action susceptibility, and the residual solver are local response, base response, and computation for one canonical fiber problem.

This response statement is local in the perturbation. A global perturbed minimum requires separate coercivity or growth control on the Δ_i .

Proof. At $(\Sigma_*, 0)$, the derivative in Σ of the stationarity equation is the invertible operator H_B . The implicit-function theorem therefore gives the unique nearby critical branch; continuity of its Hessian makes it a strict local minimum after shrinking the u -neighborhood. Differentiating stationarity and then the local value branch gives equation (20). The covariance law makes this branch independent of complement coordinates. Differentiation with respect to B uses the same stationarity equation with $E_{\Sigma B}$ in place of G_B . $\square$

5 Global SPD Action Geometry

We now realize the abstract reduction globally. Let

$$\mathcal{P}_d = \{P \in \mathbb{S}_{++}^d : \det P = 1\}, \quad r(P) = d_{\text{AI}}(I, P) = \|\log P\|_F.$$

Fix a full-column-rank $A \in \mathbb{R}^{d \times m}$, with $1 \leq m < d$ and $n = d - m$. The visible action map and its admissible base are

$$\pi_A : \mathcal{P}_d \rightarrow \mathbb{R}^{d \times m}, \quad \pi_A(P) = PA, \quad \mathcal{B}_A = \{B : A^\top B \in \mathbb{S}_{++}^m\}. \quad (21)$$

The condition $A^\top B \in \mathbb{S}_{++}^m$ includes symmetry and is necessary and sufficient for an SPD solution of $PA = B$.

Choose a thin QR factorization and orthogonal completion

$$A = UC, \quad U^\top U = I_m, \quad Q = [U, V] \in O(d).$$

For $B \in \mathcal{B}_A$, define

$$M(B) = C^{-\top} A^\top B C^{-1}, \quad N(B) = V^\top B C^{-1}, \quad (22)$$

$$T(B) = N(B) M(B)^{-1}, \quad L(B) = \begin{pmatrix} I_m & 0 \\ T(B) & I_n \end{pmatrix}, \quad (23)$$

$$\rho(B) = (\det M(B))^{-1/n}, \quad \mathcal{P}_n = \{\Sigma \in \mathbb{S}_{++}^n : \det \Sigma = 1\}. \quad (24)$$

Theorem 5 (Global action bundle and canonical controller). *The map*

$$\Phi_A : \mathcal{B}_A \times \mathcal{P}_n \rightarrow \mathcal{P}_d, \quad \Phi_A(B, \Sigma) = QL(B) \begin{pmatrix} M(B) & 0 \\ 0 & \rho(B)\Sigma \end{pmatrix} L(B)^\top Q^\top \quad (25)$$

is a global real-analytic diffeomorphism satisfying $\pi_A(\Phi_A(B, \Sigma)) = B$. Consequently,

$$\mathcal{P}_d \cong \mathcal{B}_A \times \mathcal{P}_{d-m}, \quad \mathcal{B}_A \cong \mathbb{S}_{++}^m \times \mathbb{R}^{n \times m}. \quad (26)$$

For every B , the fiber

$$\mathcal{F}_A(B) = \{P \in \mathcal{P}_d : PA = B\}$$

is nonempty, closed, totally geodesic, and AIRM-isometric to $\mathcal{P}_n$.

Every fiber has a unique nearest controller

$$s_A(B) = \arg \min_{P \in \mathcal{F}_A(B)} r(P), \quad \mathfrak{C}_A(B) = r(s_A(B)). \quad (27)$$

The section s_A and squared cost $\mathfrak{C}_A^2$ are real analytic. They are intrinsic under action-basis changes and orthogonal ambient changes:

$$s_{AD}(BD) = s_A(B), \quad s_{RA}(RB) = R s_A(B) R^\top \quad (28)$$

for $D \in \text{GL}_m$ and $R \in O(d)$.

The identity is a reference controller, not an essential restriction. For any $P_{\text{ref}} \in \mathcal{P}_d$, congruence by $P_{\text{ref}}^{-1/2}$ sends the nearest-completion problem from P_{ref} to the identity-reference problem with transformed action $\tilde{A} = P_{\text{ref}}^{1/2} A$ and $\tilde{B} = P_{\text{ref}}^{-1/2} B$. Thus the bundle and solver apply to an arbitrary determinant-one baseline without changing their geometry.

The proof responsibility in this theorem is worth separating. General Hadamard and SPD-submanifold theory supplies the unique nearest point after a fiber has been shown closed and totally geodesic Lim (2004); Tumpach & Larotonda (2024). The action-specific results are the explicit global trivialization Φ_A , the identification and total geodesy of every $PA = B$ fiber, analytic dependence on B , the residual equation below, and the solver and active reduction in section 6. Projection uniqueness by itself is not claimed as new.

The action constraint gives a complete inverse-Hessian multi-secant instance of the abstract theory. For $Y, S \in \mathbb{R}^{d \times m}$, take

$$\mathcal{X} = \mathcal{P}_d, \quad \pi_Y(P) = PY, \quad E(P) = \frac{1}{2} d_{\text{AI}}(I, P)^2, \quad y = S.$$

Then $Y^\top S \in \mathbb{S}_{++}^m$ is exactly the admissibility condition and the canonical lift $s_E(S)$ is $s_Y(S)$ from equation (27). Thus the theorem identifies every hidden SPD completion and selects the AIRM least-deformation controller for an admissible multi-secant system. Classical quasi-Newton updates select by other norms or divergences Dennis & Schnabel (1979); Fletcher (1991); Kanamori & Ohara (2013); the global bundle, total-geodesy, AIRM certificate, and active reduction are the additional objects established here.

The theorem gives a global hidden coordinate Σ . It also supplies a residual characterization and an a posteriori certificate. For fixed B , abbreviate $M = M(B)$, $L = L(B)$, and $\rho = \rho(B)$, and set

$$R_0 = L^{-1} L^{-\top}, \quad Q_\Sigma = \text{diag}(M, \rho \Sigma), \quad \mathcal{R}(\Sigma) = \log(Q_\Sigma^{-1/2} R_0 Q_\Sigma^{-1/2}). \quad (29)$$

If $X^\circ = X - \text{tr}(X)I_n/n$, the canonical hidden coordinate is the unique solution of

$$\mathcal{R}_{22}(\Sigma_\star)^\circ = 0. \quad (30)$$

For every trial $\Sigma \in \mathcal{P}_n$,

$$d_{\text{AI}}(\Sigma, \Sigma_\star) \leq \|\mathcal{R}_{22}(\Sigma)^\circ\|_F, \quad (31)$$

$$0 \leq f_B(\Sigma) - f_B(\Sigma_\star) \leq \frac{1}{2} \|\mathcal{R}_{22}(\Sigma)^\circ\|_F^2, \quad (32)$$

where $f_B(\Sigma) = \frac{1}{2} d_{\text{AI}}(R_0, Q_\Sigma)^2$.

5.1 Active reduction

Let

$$W = \text{span}(\text{col } A, \text{col } B), \quad r = \dim W \leq 2m, \quad q = d - r.$$

If $q > 0$, the canonical controller has the form

$$s_A(B) = H_\star \oplus (\det H_\star)^{-1/q} I_q, \quad (33)$$

where $H_\star \in \text{SPD}(W)$ uniquely minimizes

$$\min_{\substack{H \in \text{SPD}(W) \\ HA=B}} \left\{ \|\log H\|_F^2 + \frac{(\log \det H)^2}{q} \right\}. \quad (34)$$

Thus the matrix variable is supported on an ambient subspace of dimension $r \leq 2m$, rather than d . It still has $O(r^2)$ scalar degrees of freedom; the claim is an active-subspace reduction, not a $2m$ -variable parameterization. The next section turns the normal equation into a globally convergent residual-gradient iteration and proves that this active symmetry is preserved at every step when it is present in the initializer; the identity initializer has the required symmetry.

5.2 Scale and reference

The determinant constraint fixes unit-volume shape; it does not create a dynamical gauge, because rescaling changes PA and the update time scale. Dropping the constraint replaces $\mathcal{P}_n$ by $\mathbb{S}_{++}^n$. The complement volume keeps the admissible base $\mathcal{B}_A$ unchanged, while the feasible fiber and canonical section change. The compression $P \mapsto A^\top PA$ has a different fiber (Li, 2026); here the complete action PA is fixed.

6 A Global Reduced Solver with Posterior Certificates

We first isolate the computation principle behind canonical reduction. This also clarifies which part is general Hadamard optimization and which part is specific to the SPD action bundle.

Theorem 6 (Canonical strongly-convex fiber engine). *Let $\mathcal{C}$ be a closed geodesically convex totally geodesic submanifold of a finite-dimensional Hadamard manifold. Suppose $F : \mathcal{C} \rightarrow \mathbb{R}$ is C^2 , coercive, and μ -strongly geodesically convex. If*

$$\text{Hess } F \preceq L_0 g \quad \text{on } \{x : F(x) \leq F(x_0)\},$$

then F has a unique minimizer $x_\star$, every complete update segment of

$$x_{k+1} = \text{Exp}_{x_k}(-L_0^{-1} \text{grad } F(x_k)) \quad (35)$$

stays in the current sublevel set, and, with $q_{\mu,L} = 1 - \mu/L_0$,

$$\Delta_k \leq q_{\mu,L}^k \Delta_0, \quad d(x_k, x_\star) \leq \sqrt{\frac{2\Delta_0}{\mu}} q_{\mu,L}^{k/2}, \quad (36)$$

$$\|\text{grad } F(x_k)\| \leq \sqrt{2L_0\Delta_0} q_{\mu,L}^{k/2}. \quad (37)$$

At every $x \in \mathcal{C}$,

$$d(x, x_\star) \leq \frac{\|\text{grad } F(x)\|}{\mu}, \quad 0 \leq F(x) - F(x_\star) \leq \frac{\|\text{grad } F(x)\|^2}{2\mu}. \quad (38)$$

The constants in equation 38 are sharp.

The first-return sublevel statement is important: it uses a Hessian bound on the current sublevel itself, rather than first enlarging its radius to cover a putative update. For squared distance on a Hadamard manifold with $-\kappa \leq \text{sec} \leq 0$, Hessian comparison supplies the sharp radius-only majorant

$$L(D) = \psi(\sqrt{\kappa}D), \quad \psi(s) = \begin{cases} s \coth s, & s > 0, \\ 1, & s = 0. \end{cases} \quad (39)$$

The general descent ingredients are classical; the theorem records the exact form needed to connect the canonical section to the action-specific result.

6.1 Explicit SPD residual and nonasymptotic complexity

For fixed $B \in \mathcal{B}_A$, retain $R_0, \mathcal{Q}_\Sigma, f_B$, and $\mathcal{R}(\Sigma)$ from section 5, and define

$$W_B(\Sigma) = \mathcal{R}_{22}(\Sigma)^\circ. \quad (40)$$

This is the exact whitened negative gradient:

$$-\text{grad } f_B(\Sigma) = \Sigma^{1/2} W_B(\Sigma) \Sigma^{1/2}, \quad \|\text{grad } f_B(\Sigma)\|_\Sigma = \|W_B(\Sigma)\|_F. \quad (41)$$

Given any $\Sigma_0 \in \mathcal{P}_n$, put

$$D_0 = d_{\text{AI}}(R_0, \mathcal{Q}_{\Sigma_0}), \quad L_0 = \psi(D_0/\sqrt{2}), \quad q_0 = 1 - L_0^{-1}. \quad (42)$$

If $D_0 = 0$, the initial controller is canonical. Otherwise define

$$\chi_0 = -\log q_0 = -\log(1 - L_0^{-1}) \quad (43)$$

and run

$$\Sigma_{k+1} = \Sigma_k^{1/2} \exp(L_0^{-1} W_B(\Sigma_k)) \Sigma_k^{1/2}. \quad (44)$$

Theorem 7 (Global action solver and posterior certificates). *Let A have full column rank, let $B \in \mathcal{B}_A$, and let $\Sigma_\star$ be the hidden coordinate of $s_A(B)$. From every $\Sigma_0 \in \mathcal{P}_n$, equation 44 preserves determinant one and satisfies*

$$f_B(\Sigma_{k+1}) \leq f_B(\Sigma_k) - \frac{1}{2L_0} \|W_B(\Sigma_k)\|_F^2, \quad (45)$$

$$f_B(\Sigma_k) - f_B(\Sigma_\star) \leq \frac{D_0^2}{2} q_0^k, \quad (46)$$

$$d_{\text{AI}}(\Sigma_k, \Sigma_\star) = d_{\text{AI}}(\Phi_A(B, \Sigma_k), s_A(B)) \leq D_0 q_0^{k/2}, \quad (47)$$

$$\|W_B(\Sigma_k)\|_F \leq \sqrt{L_0} D_0 q_0^{k/2}. \quad (48)$$

The observable residual gives the sharp posterior certificates

$$d_{\text{AI}}(\Phi_A(B, \Sigma_k), s_A(B)) \leq \|W_B(\Sigma_k)\|_F, \quad (49)$$

$$0 \leq f_B(\Sigma_k) - f_B(\Sigma_\star) \leq \frac{1}{2} \|W_B(\Sigma_k)\|_F^2. \quad (50)$$

For $D_0 > 0$, value, controller, and residual tolerances are guaranteed, respectively, after

$$k \geq \frac{1}{\chi_0} \left[\log \frac{D_0^2}{2\varepsilon_f} \right]_+, \quad (51)$$

$$k \geq \frac{2}{\chi_0} \left[\log \frac{D_0}{\varepsilon_x} \right]_+, \quad (52)$$

$$k \geq \frac{2}{\chi_0} \left[\log \frac{\sqrt{L_0} D_0}{\varepsilon_r} \right]_+ \quad (53)$$

iterations, with ceilings understood.

The current-sublevel curvature comparison uses $L_0 = \psi(D_0/\sqrt{2})$, the sharp dimension-uniform majorant. No radius-doubling argument is needed. The denominators above invert the discrete factor q_0 exactly; $1/\chi_0 \leq L_0$ recovers the simpler exponential estimates.

6.2 Active realization and spectral-arithmetic cost

The operation counts below use a spectral-arithmetic model: dense factorizations and matrix functions on an $r \times r$ symmetric matrix cost $O(r^3)$, with scalar elementary functions counted at unit cost. These are algebraic operation counts, not bit-complexity or floating-point error bounds.

Let

$$\mathcal{W}_A = \text{span}(\text{col } A, \text{col } B), \quad r = \dim \mathcal{W}_A \leq 2m, \quad s = r - m, \quad q = d - r.$$

The active-adapted construction in equation (87) represents the hidden state by $\Sigma_a \oplus \beta I_q$, with $\det \Sigma_a \beta^q = 1$, and the full residual norm by $\|W_a\|_F^2 + qw_c^2$. All complement terms are omitted when $q = 0$. This family is invariant under the exact update. A general $\Sigma_0 \in \mathcal{P}_{d-m}$ need not lie in it, so Theorem 7 gives arbitrary-start convergence while the operation count below applies to active-compatible starts. The identity is the default such start.

Corollary 2 (Active canonical solver). *Let $\Sigma_{a,0} \in \mathbb{S}_{++}^s$ and, when present, $\beta_0 > 0$ satisfy $\det \Sigma_{a,0} \beta_0^q = 1$; the identity choice $\Sigma_{a,0} = I_s$, $\beta_0 = 1$ is admissible. The updates*

$$\Sigma_{a,k+1} = \Sigma_{a,k}^{1/2} e^{W_{a,k}/L_0} \Sigma_{a,k}^{1/2}, \quad \beta_{k+1} = \beta_k e^{w_{c,k}/L_0} \quad (54)$$

are exactly the full iteration equation 44. They preserve $P_k A = B$, $P_k \succ 0$, and $\det P_k = 1$, while the stopping test $\|W_{a,k}\|_F^2 + qw_{c,k}^2 \leq \varepsilon^2$ implies controller error at most ε .

Preprocessing costs $O(dm^2 + r^3)$ spectral-arithmetic operations and $O(dm + r^2)$ storage. Each iteration costs $O(r^3)$. For $D_0 > 0$, the total spectral-arithmetic cost is

$$O\left(dm^2 + r^3 \left\{1 + \frac{1}{\chi_0} \left[\log \frac{\sqrt{L_0} D_0}{\varepsilon}\right]_+\right\}\right), \quad r \leq 2m. \quad (55)$$

The implicit controller application costs $O(dr + r^2)$. The same formulas cover $q = 0$; if $s = 0$, the solver has no variable and the canonical controller is

$$s_A(B) = EME^\top + \rho(I - EE^\top).$$

6.3 Conditional inexact residuals

Finite-precision propagation requires certified numerical information that is separate from the geometric solver theorem. Assume a matrix-function routine supplies a rigorous initial radius majorant $\widehat{D}_0 \geq D_0$, a symmetric trace-free direction $\widetilde{W}_k = W_B(\Sigma_k) + E_k$, and an error majorant $b_k \geq \|E_k\|_F$. Such bounds require a certified matrix-function implementation; they are not produced by the analysis below Higham (2008).

Fix $0 \leq \delta < 1$ and require at every accepted step

$$b_k \leq \frac{\delta}{1 + \delta} \|\widetilde{W}_k\|_F. \quad (56)$$

This observable test implies

$$\|E_k\|_F \leq \delta \|W_B(\Sigma_k)\|_F. \quad (57)$$

Define

$$\widehat{L}_0 = \psi(\widehat{D}_0/\sqrt{2}), \quad c_\delta = \left(\frac{1 - \delta}{1 + \delta}\right)^2, \quad \widehat{q}_\delta = 1 - \frac{c_\delta}{\widehat{L}_0}, \quad \eta = \frac{1 - \delta}{\widehat{L}_0(1 + \delta)^2}. \quad (58)$$

Corollary 3 (Certified inexact residual-oracle propagation). *If the supplied majorants satisfy the preceding conditions, the update*

$$\Sigma_{k+1} = \Sigma_k^{1/2} \exp(\eta \widetilde{W}_k) \Sigma_k^{1/2}$$

has nonincreasing objective radius and

$$f_B(\Sigma_k) - f_B(\Sigma_\star) \leq \frac{\hat{D}_0^2}{2} \hat{q}_\delta^k, \quad (59)$$

$$d_{\text{AI}}(\Phi_A(B, \Sigma_k), s_A(B)) \leq \hat{D}_0 \hat{q}_\delta^{k/2}, \quad (60)$$

$$\|\widetilde{W}_k\|_F \leq \frac{(1+\delta)^2}{1-\delta} \sqrt{\hat{L}_0 \hat{D}_0} \hat{q}_\delta^{k/2}. \quad (61)$$

The observed residual also certifies

$$d_{\text{AI}}(\Phi_A(B, \Sigma_k), s_A(B)) \leq \frac{\|\widetilde{W}_k\|_F}{1-\delta}, \quad f_B(\Sigma_k) - f_B(\Sigma_\star) \leq \frac{\|\widetilde{W}_k\|_F^2}{2(1-\delta)^2}. \quad (62)$$

This corollary certifies propagation of error in the residual direction $\widetilde{W}_k$ when the accepted update is the exact Riemannian exponential of that direction. It does not cover roundoff in the exponential update, factorizations, active-basis construction, determinant maintenance, or stored iterates, and it is not a bit-complexity theorem for matrix logarithms, roots, or exponentials. End-to-end finite-precision certification requires those additional errors to be enclosed by the implementation.

7 Controlled Numerical Validation

The main results are analytic. This section tests the computational identities and certificates on deterministic synthetic instances, with no learned model or external data. In orthonormal action coordinates we set

$$A = \begin{pmatrix} I_m \\ 0 \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} M \\ N_a \\ 0 \end{pmatrix}, \quad M \succ 0,$$

where $N_a \in \mathbb{R}^{s \times m}$, so the active dimension is $r = m + s$. The generator forms R_0 , ρ , and both residuals from the formulas in section 5 and equation (87). Matrix roots, logarithms, and exponentials use symmetric eigendecompositions. All reported distributions use the fixed seeds 7, 19, 43; the complete configuration and raw results accompany the source.

Certificates and exact active realization. For $d = 24$, $m = s = 4$, we computed a reference solution to residual 10^{-13} , then ran both the full and active iterations from the identity until the full residual was at most 10^{-5} . Across all recorded iterates, the largest ratios

$$\frac{d_{\text{AI}}(\Sigma_k, \Sigma_\star)}{\|W_B(\Sigma_k)\|_F} \quad \text{and} \quad \frac{f_B(\Sigma_k) - f_B(\Sigma_\star)}{\frac{1}{2} \|W_B(\Sigma_k)\|_F^2}$$

were 0.972 and 0.971, respectively, consistent with equations (49) and (50). The maximum ratios are conservative: the reference certificate contributes 10^{-13} to the numerator of the distance ratio and at most 5×10^{-27} to the objective-gap numerator. The maximum relative difference between the reconstructed active iterate and the full iterate was 2.1×10^{-15} . Relative action infeasibility stayed below 1.1×10^{-16} , and absolute log-determinant drift below 1.8×10^{-14} . These last quantities measure floating-point drift; they are not interval certificates.

Dimension and boundary sensitivity. We fixed $m = s = 4$ and used ambient dimensions 32, 64, 96, 128. The full step acts on a dense matrix whose order grows with d , whereas the active step keeps an order-eight spectral kernel. At $d = 128$, the observed median one-step times were 8.57 ms and 0.307 ms. The median of the 21 paired full/active time ratios was 26.9 in this reference Python implementation. The corresponding storage proxies d^2 and $dm + r^2$ are 16384 and 576 scalar entries.

To approach the feasibility boundary, we varied $\lambda_{\min}(A^\top B) = \lambda_{\min}(M)$ from 1 to 10^{-3} while holding dimensions and random directions fixed. Median D_0 grew from 0.994 to 13.98, median L_0 from 1.159 to

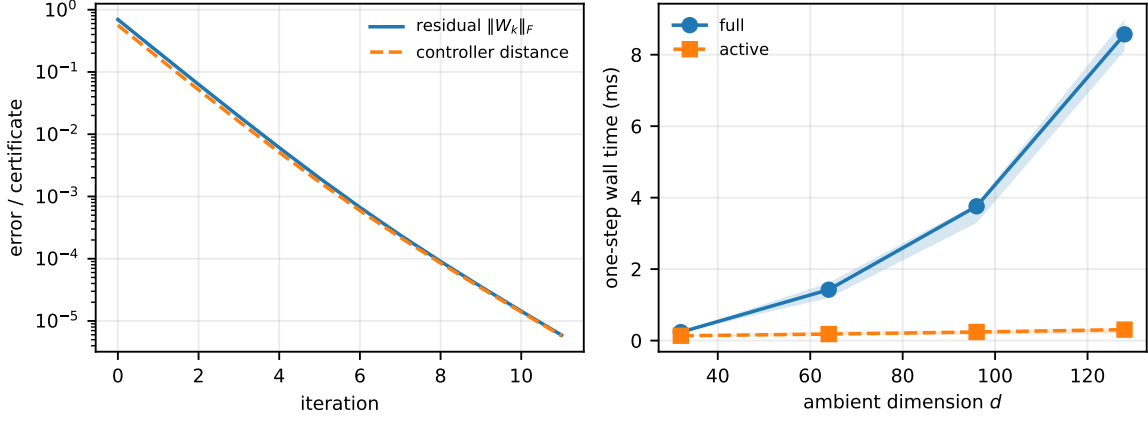


Figure 1: Left: residual certificate and reference controller distance for one fixed synthetic instance; the other seeds give the same qualitative decay. Right: median wall time for one full and active step, with interquartile bands over three seeds and seven repetitions after an unmeasured warm-up; measurement order is alternated and BLAS uses one thread. Here $m = s = 4$, so the active spectral kernel remains $r = 8$ while d grows. Timings are implementation and hardware dependent; the invariant result is the kernel dimension in Corollary 2.

9.885, and the median iterations needed for residual 10^{-8} from 8 to 126 (the range over all instances was 8–143). Thus the experiment exposes the rate constant’s sensitivity before feasibility is lost; it does not extrapolate the theorem to singular $A^\top B$.

Same-objective baseline and nested recovery. We also compared the explicit $1/L_0$ step with a Riemannian gradient method using Armijo backtracking on the same hidden-coordinate objective and from the same identity start. At residual tolerance 10^{-8} , median iteration counts over the three seeds were 17 and 11, while median objective evaluations were 18 and 23, respectively. Thus line search used fewer accepted steps but more objective evaluations; the explicit step supplies its rate and admissibility without backtracking. This comparison isolates step selection and does not compare against methods with different feasible models.

Finally, for random 8×8 Hessians we used cumulative ranks 2, 4, 7 and stopped each inner projection at residual 10^{-3} . Across three seeds, the largest actual-inner-error/certificate ratio was 0.983, and the largest left/right ratio in equation (72) was $1 + 9.3 \times 10^{-15}$. The rank-seven output recovered $P_\star$ within 1.3×10^{-14} AIRM distance; relative cumulative-action error stayed below 2.4×10^{-15} . This test evaluates the exact-feasibility model in floating point and reports the resulting feasibility drift separately.

These experiments test the exact formulas in standard floating point. They do not instantiate the certified residual-oracle model of Corollary 3; an end-to-end implementation of that model would require rigorous enclosures for every matrix function and update.

8 Canonical Multi-Secant Recovery

The posterior-certified exact solver computes one canonical action projection. Nested action fibers turn those projections into a finite inverse-shape recovery law. Let $H \in \mathbb{S}_{++}^d$ be a fixed quadratic Hessian and define its determinant-one inverse shape

$$P_\star = c_H H^{-1}, \quad c_H = (\det H)^{1/d}. \quad (63)$$

The scalar c_H is assumed known, or supplied by an independent scalar gauge. This information requirement is part of the theorem.

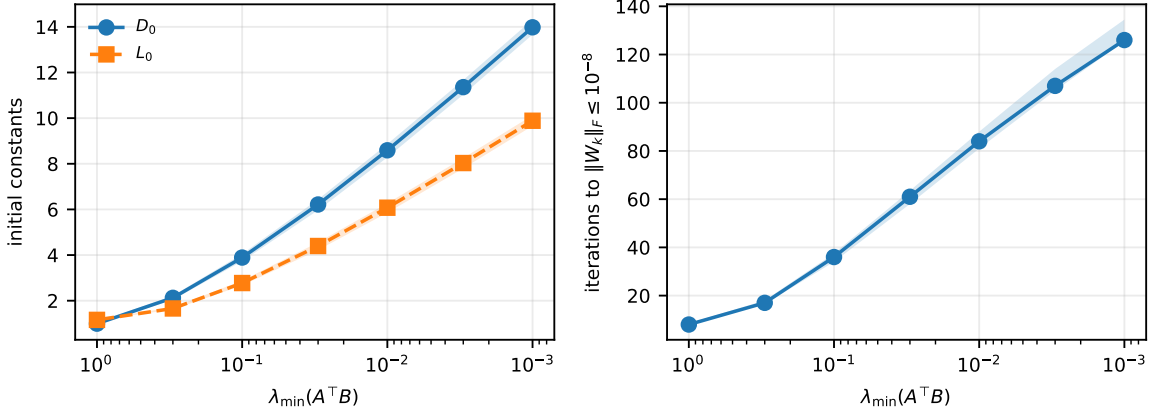


Figure 2: Sensitivity as $A^T B$ approaches the positive-definite boundary. Curves show medians and shaded interquartile ranges over the three fixed seeds. Both the analytic radius majorant and the observed iteration count deteriorate while every tested instance remains feasible.

Let $S^{(1)}, \dots, S^{(K)}$ have strictly nested column spaces, delete dependent columns, and write $r_k = \text{rank } S^{(k)}$. Define the cumulative shape-normalized actions

$$A_k = H S^{(k)}, \quad B_k = c_H S^{(k)}, \quad \mathcal{C}_k = \{P \in \mathcal{P}_d : P A_k = B_k\}. \quad (64)$$

Then $P_\star \in \mathcal{C}_k$ and $\mathcal{C}_{k+1} \subset \mathcal{C}_k$. Starting from any $P_0 \in \mathcal{P}_d$, define

$$P_k = \arg \min_{P \in \mathcal{C}_k} d_{\text{AI}}(P_{k-1}, P). \quad (65)$$

Theorem 8 (Nested canonical multi-secant identification). *For the exact outer projections in equation 65, every update exists uniquely and preserves all earlier cumulative secants. For every k ,*

$$\boxed{d_{\text{AI}}(P_k, P_\star)^2 + d_{\text{AI}}(P_{k-1}, P_k)^2 \leq d_{\text{AI}}(P_{k-1}, P_\star)^2.} \quad (66)$$

Consequently,

$$\sum_{j=1}^k d_{\text{AI}}(P_{j-1}, P_j)^2 \leq d_{\text{AI}}(P_0, P_\star)^2 - d_{\text{AI}}(P_k, P_\star)^2, \quad (67)$$

$$\sum_{j=1}^k d_{\text{AI}}(P_{j-1}, P_j) \leq \sqrt{k} d_{\text{AI}}(P_0, P_\star). \quad (68)$$

The determinant-one identification threshold is sharp:

$$r_K = d - 1 \implies P_K = P_\star, \quad (69)$$

whereas any cumulative rank $r \leq d - 2$ leaves infinitely many feasible determinant-one SPD completions. If each pre-final round contributes exactly b new independent probe directions and the final round contributes the remainder, exact recovery occurs after

$$K = \left\lceil \frac{d-1}{b} \right\rceil \quad (70)$$

rounds. Each exact outer projection reduces by AIRM congruence to the identity-reference problem in Theorem 7; the inner iteration converges globally to it and supplies a residual certificate for a finite approximation.

The outer decrease remains quantitative when each projection is stopped after finitely many certified inner iterations. Set $\tilde{P}_0 = P_0$. At round k , let

$$Q_k = \text{Proj}_{\mathcal{C}_k}(\tilde{P}_{k-1})$$

be the exact projection used only for analysis, and let $\tilde{P}_k \in \mathcal{C}_k$ be the feasible inner output. This theorem uses the exact-feasibility model: action-chart inner iterates lie in $\mathcal{C}_k$ in exact arithmetic. Floating-point feasibility error is a separate perturbation and is not included in ε_k .

Theorem 9 (Residual-controlled finite inner projections). *Suppose a posterior inner certificate gives*

$$d_{\text{AI}}(\tilde{P}_k, Q_k) \leq \varepsilon_k. \quad (71)$$

Then, with $D_k = d_{\text{AI}}(\tilde{P}_k, P_\star)$,

$$\begin{aligned} d_{\text{AI}}(\tilde{P}_{k-1}, \tilde{P}_k)^2 + D_k^2 \\ \leq D_{k-1}^2 + 2\sqrt{2}\varepsilon_k D_{k-1} + 2\varepsilon_k^2, \end{aligned} \quad (72)$$

$$D_k \leq D_{k-1} + \varepsilon_k \leq D_0 + \sum_{j=1}^k \varepsilon_j. \quad (73)$$

Consequently,

$$\sum_{j=1}^k d_{\text{AI}}(\tilde{P}_{j-1}, \tilde{P}_j)^2 \leq D_0^2 - D_k^2 + \sum_{j=1}^k \left(2\sqrt{2}\varepsilon_j D_{j-1} + 2\varepsilon_j^2 \right). \quad (74)$$

All earlier cumulative actions remain exact because $\tilde{P}_k \in \mathcal{C}_k \subseteq \mathcal{C}_j$ for $j < k$. After congruence to the identity-reference inner problem, equation (49) permits the observable choice $\varepsilon_k = \|W_{B,k}(\tilde{\Sigma}_k)\|_F$; under the residual-oracle model, equation (62) supplies the corresponding enlarged bound. If $r_K = d - 1$, then $\mathcal{C}_K = \{P_\star\}$, so every feasible inner output at the final round equals $P_\star$ exactly.

The known scalar gauge cannot be dropped. Ordinary Hessian-vector secants in $d - 1$ directions do not determine the inverse shape. In dimension two,

$$H_t = \text{diag}(1, t), \quad s = e_1$$

gives the same observation $H_t s = e_1$ for every $t > 0$, while

$$(\det H_t)^{1/2} H_t^{-1} = \text{diag}(\sqrt{t}, 1/\sqrt{t})$$

varies with t . Thus equation 69 is a sharp theorem for shape-normalized exact actions, rather than a claim of scale-free identification from ordinary secants. Acquisition of c_H is external to the action oracle and is not claimed to be cheaper than an additional probe.

9 Conclusion and Extensions

We studied affine-invariant least-deformation completion from the exact action $PA = B$. The global bundle theorem identifies every determinant-one feasible fiber and its analytic nearest-controller section. Total geodesy and the explicit logarithmic residual turn this geometric selection into a globally linearly convergent reduced solver with posterior certificates. From the identity, or another active-compatible hidden state, the active realization confines all dense matrix functions to dimension $r \leq 2m$; arbitrary hidden starts retain convergence without this operation bound. Nested action fibers further yield exact and residual-controlled approximate Pythagorean decreases and the sharp rank threshold for normalized inverse-Hessian recovery.

Variational reduction unifies these conclusions. Its vertical Hessian governs both response and computation: in Corollary 1, H_B^{-1} maps mechanism perturbations to interaction curvature and visible-action perturbations to

canonical-section susceptibility. For the unperturbed completion problem, the same fiber geometry produces the optimization residual and its certificates. The resulting chain is

$$\begin{aligned} \text{reduction} &\longrightarrow \text{vertical response} \longrightarrow \text{action bundle} \\ &\longrightarrow \text{posterior-certified active solver.} \end{aligned}$$

The controlled experiments reproduce the posterior inequalities, exact full-active trajectory equivalence to numerical precision, and the fixed-size active spectral kernel as d grows. They also show increasing initial radius, curvature majorant, and iteration count as $A^\top B$ approaches the positive-definite boundary. This is a conditioning diagnostic, not an analytic boundary-rate theorem.

Several extensions arise directly from the formulation. An end-to-end finite-precision implementation must supplement the residual-oracle theorem with certified exponential updates, factorizations, active bases, determinant maintenance, and stored iterates; combining those enclosures with the present spectral-arithmetic count would yield a bit-complexity result. The recovery law can be extended by treating the determinant gauge as an additional unknown or observation. Structured controller families replace the totally geodesic full-SPD fiber by a restricted feasible set, while noisy actions replace exact completion by statistical or robust projection. These variants provide the route from exact canonical completion to implementable adaptive preconditioners.

Data and Code Availability

The deterministic synthetic generator, full and active reference implementations, fixed configuration, raw CSV results, and figure-generation script are included in the source archive under `experiments/`. They use no external datasets. Seeds, dimensions, tolerances, repetitions, and software requirements are recorded in `experiments/config.json` and `experiments/README.md`. The code evaluates the exact-arithmetic formulas with standard floating-point eigendecompositions; it is numerical validation, not a certified matrix-function package. The residual-oracle result still assumes rigorous direction-error and initial-radius majorants satisfying equation (56), and its implementation boundary is stated after Corollary 3.

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A Proofs for Variational Reduction and Response

A.1 Algebraic reduction

Proof of Theorem 1. Fix $z \in \mathcal{Z}$. Partitioning the composite fiber by its intermediate visible value gives

$$\begin{aligned} ((\rho \circ \pi)_! E)(z) &= \inf_{x: \rho(\pi(x))=z} E(x) \\ &= \inf_{y: \rho(y)=z} \inf_{x: \pi(x)=y} E(x) \\ &= (\rho_!(\pi_! E))(z). \end{aligned}$$

Likewise,

$$\begin{aligned} \inf_x \{E(x) + \ell(\pi(x))\} &= \inf_y \inf_{\pi(x)=y} \{E(x) + \ell(y)\} \\ &= \inf_y \{(\pi_! E)(y) + \ell(y)\}. \end{aligned}$$

The base-potential law follows because $\varphi(\pi(x)) = \varphi(y)$ is constant on each fiber. The constraint law follows from the definition of the indicator energy. No attainment assumption is used. $\square$

A.2 Existence and uniqueness of the canonical lift

Proof of Theorem 2. Fix y with finite reduced energy. Coercivity places every minimizing sequence in a bounded sublevel set. Properness makes its closure compact, and lower semicontinuity gives a minimizer in the closed fiber.

If $x_0 \neq x_1$ were two minimizers, their fiber geodesic would satisfy

$$E(x_0 \#_{1/2} x_1) \leq \frac{1}{2}E(x_0) + \frac{1}{2}E(x_1) - \frac{\mu}{8}d(x_0, x_1)^2,$$

contradicting minimality. Thus the minimizer is unique.

For equivariance, $gs_E(y)$ lies in the fiber over gy and has the same energy as $s_E(y)$. Uniqueness on that fiber gives $s_E(gy) = gs_E(y)$. $\square$

A.3 Smooth minimizer response

Proof of Theorem 3. We first show continuity of the global minimizer section. Let $z_k \rightarrow z$. Choose one fixed trial point near $\sigma_*(z)$. Continuity of E uniformly bounds the optimal values for all sufficiently large k . Local uniform inf-compactness therefore places $\sigma_*(z_k)$ in a common relatively compact set. Every convergent subsequence has a limit $\bar{\sigma}$. Optimality and continuity give, for every trial σ ,

$$E(z, \bar{\sigma}) = \lim E(z_k, \sigma_*(z_k)) \leq \lim E(z_k, \sigma) = E(z, \sigma).$$

The minimizer at z is unique, so $\bar{\sigma} = \sigma_*(z)$. Every subsequence has the same limit, proving continuity.

The first-order condition is

$$E_\sigma(z, \sigma_*(z)) = 0. \tag{75}$$

Its derivative with respect to σ is the invertible matrix H . The implicit-function theorem gives a local C^r critical section. The continuous global minimizer section agrees with it near every point and is therefore C^r . Differentiating equation (75) gives

$$E_{\sigma z} + H D_z \sigma_* = 0,$$

which proves equation (9). The chain rule gives

$$D_z \bar{E} = E_z + E_\sigma D_z \sigma_* = E_z,$$

and a second derivative gives

$$D_{zz}^2 \bar{E} = E_{zz} + E_{z\sigma} D_z \sigma_* = E_{zz} - E_{z\sigma} H^{-1} E_{\sigma z}.$$

For nested variables $\sigma = (\alpha, \beta)$, let q be the quadratic form defined by the complete Hessian at the optimum. Positive definiteness of the joint hidden block makes both quadratic minimizations unique. Algebraic reduction gives

$$\inf_{\delta\alpha, \delta\beta} q(\delta z, \delta\alpha, \delta\beta) = \inf_{\delta\alpha} \left(\inf_{\delta\beta} q(\delta z, \delta\alpha, \delta\beta) \right).$$

The matrix of a partially minimized quadratic form is the corresponding Schur complement. Equality for every δz proves equation (12). $\square$

A.4 Interaction curvature and finite contrasts

Proof of Theorem 4. For equation (13),

$$E_{uu} = 0, \quad E_{\sigma u} = G.$$

The u -block of equation (11) is therefore

$$D_{uu}^2 \bar{E} = -G^* H^{-1} G.$$

For every $a \in \mathbb{R}^p$,

$$\langle a, D_{uu}^2 \bar{E} a \rangle = -\langle Ga, H^{-1} Ga \rangle \leq 0,$$

because $H^{-1} \succ 0$. Expanding in coordinate vectors gives equation (16). A mixed entry is zero exactly when the two corresponding covectors are orthogonal under the inner product induced by H^{-1} . Finally, without differentiability,

$$\bar{E}(y, u) = \inf_{\sigma} \left(E_0(y, \sigma) + \sum_i u_i \Delta_i(y, \sigma) \right)$$

is a pointwise infimum of affine functions of u , hence concave. $\square$

A.5 Finite contrasts and order of reduction

Corollary 4 (Factorial finite differences). *Assume $[0, 1]^p \subset \mathcal{U}$. For $T \subseteq [p]$, define*

$$\delta_T \bar{E} = \sum_{B \subseteq T} (-1)^{|T|-|B|} \bar{E}(y, \mathbf{1}_B). \quad (76)$$

If $\bar{E} \in C^{|T|}$, then

$$\delta_T \bar{E} = \int_{[0,1]^T} \partial_T^{|T|} \bar{E}(u_T, 0_{T^c}) du_T. \quad (77)$$

Proof. For a single coordinate, the fundamental theorem of calculus gives

$$\bar{E}(e_i) - \bar{E}(0) = \int_0^1 \partial_{u_i} \bar{E}(te_i) dt.$$

Apply the same identity successively to each coordinate in T . The product of difference operators expands as

$$\prod_{i \in T} (\mathbb{T}_i - I) \bar{E}(0) = \sum_{B \subseteq T} (-1)^{|T|-|B|} \bar{E}(\mathbf{1}_B),$$

and repeated integration gives

$$\prod_{i \in T} (\mathbb{T}_i - I) \bar{E}(0) = \int_{[0,1]^T} \partial_T^{|T|} \bar{E}(u_T, 0_{T^c}) du_T.$$

For $T = \{i, j\}$, substitute equation (16) to obtain equation (18). $\square$

Reduction need not commute with finite differencing. For

$$E_0(\sigma) = \sigma^2, \quad E_1(\sigma) = (\sigma - 1)^2,$$

both reduced minima are zero, but $\inf_{\sigma}(E_1 - E_0) = \inf_{\sigma}(1 - 2\sigma) = -\infty$. Thus hidden states must be reduced first. If every perturbation Δ_i is constant on each fiber, the minimizer is unchanged and all higher-order contrasts vanish; this is the exact commuting case.

B Proof of the Global SPD Action Bundle

Proof of Theorem 5. We proceed in seven steps.

Step 1: coordinates on the action base. Put $Y = BC^{-1}$. Since $Q = [U, V]$ is orthogonal,

$$Y = UM + VN, \quad M = U^{\top}Y, \quad N = V^{\top}Y.$$

Moreover,

$$A^{\top}B = C^{\top}MC,$$

so $A^{\top}B \in \mathbb{S}_{++}^m$ exactly when $M \in \mathbb{S}_{++}^m$. The map $B \mapsto (M, N)$ has the analytic inverse

$$B = (UM + VN)C. \tag{78}$$

Thus $\mathcal{B}_A \cong \mathbb{S}_{++}^m \times \mathbb{R}^{n \times m}$.

Step 2: every SPD completion. Suppose $P \in \mathbb{S}_{++}^d$ satisfies $PA = B$. Because $A = UC$, this is equivalent to $PU = Y$. In the Q coordinates, symmetry fixes the first block row and column:

$$\hat{P} := Q^{\top}PQ = \begin{pmatrix} M & N^{\top} \\ N & D \end{pmatrix}.$$

The Schur-complement criterion gives

$$\hat{P} \succ 0 \iff S := D - NM^{-1}N^{\top} \succ 0.$$

With $T = NM^{-1}$, every positive solution and only such a solution is

$$\hat{P} = L \begin{pmatrix} M & 0 \\ 0 & S \end{pmatrix} L^{\top}. \tag{79}$$

Since $\det L = 1$,

$$\det P = \det M \det S.$$

The determinant-one constraint is therefore equivalent to

$$S = \rho \Sigma, \quad \rho = (\det M)^{-1/n}, \quad \det \Sigma = 1,$$

which is exactly equation (25). Conversely, direct block multiplication and equation (78) give

$$\Phi_A(B, \Sigma)A = Q \begin{pmatrix} M \\ N \end{pmatrix} C = B.$$

Step 3: analytic inverse. Given $P \in \mathcal{P}_d$, recover $B = PA$, then $M(B), N(B)$, and the unique Schur complement

$$S = D - NM^{-1}N^{\top}.$$

The unique determinant-one hidden coordinate is

$$\Sigma = (\det M)^{1/n} S. \tag{80}$$

All operations are analytic on their SPD domains. Thus Φ_A is a global analytic diffeomorphism and π_A is projection onto its first factor.

Step 4: fiber geometry. Fix B . Before congruence by QL , the fiber is

$$\mathcal{C}_B = \{\text{diag}(M, \rho\Sigma) : \Sigma \in \mathcal{P}_n\}.$$

The ambient AIRM geodesic between two such points is

$$\text{diag}(M, \rho\Sigma_0) \#_t \text{diag}(M, \rho\Sigma_1) = \text{diag}(M, \rho(\Sigma_0 \#_t \Sigma_1)).$$

Because log determinant is affine along AIRM geodesics, $\det(\Sigma_0 \#_t \Sigma_1) = 1$. Hence the fiber is totally geodesic. It is closed, and block distance splitting gives

$$d_{\text{AI}}(\text{diag}(M, \rho\Sigma_0), \text{diag}(M, \rho\Sigma_1)) = d_{\text{AI}}(\Sigma_0, \Sigma_1).$$

Congruence by QL is an AIRM isometry, proving all fiber claims.

Step 5: canonical projection and analyticity. The determinant-one SPD slice is a finite-dimensional Hadamard manifold. A nonempty closed geodesically convex set has a unique metric projection, so equation (27) defines a global section.

In the global coordinates define

$$F(B, \Sigma) = \frac{1}{2} \|\log \Phi_A(B, \Sigma)\|_F^2.$$

This function is analytic. For fixed B , its hidden variable parameterizes an isometric totally geodesic fiber. Squared distance to I , restricted to that fiber, is strongly geodesically convex. At the unique critical point the vertical Hessian is positive definite. The analytic implicit-function theorem therefore gives a local analytic minimizer $\Sigma_\star(B)$; global uniqueness makes these local sections agree. Hence

$$s_A(B) = \Phi_A(B, \Sigma_\star(B))$$

and $\mathfrak{C}_A(B)^2 = 2F(B, \Sigma_\star(B))$ are analytic.

Step 6: equivariance. For $D \in \text{GL}_m$,

$$P(AD) = BD \iff PA = B,$$

so the two action fibers are identical and have the same unique projection. For $R \in O(d)$,

$$PA = B \iff (RPR^\top)(RA) = RB.$$

Orthogonal congruence preserves determinant and distance to I , so uniqueness gives equation (28).

Step 7: normal equation and certificate. Congruence invariance reduces the hidden objective to

$$f_B(\Sigma) = \frac{1}{2} d_{\text{AI}}(R_0, \mathcal{Q}_\Sigma)^2.$$

We use the convention that a function is 1-strongly geodesically convex when every constant-speed geodesic γ_t satisfies

$$f(\gamma_t) \leq (1-t)f(\gamma_0) + tf(\gamma_1) - \frac{1}{2}t(1-t)d(\gamma_0, \gamma_1)^2. \quad (81)$$

On a CAT(0) space this inequality holds for $f(\cdot) = \frac{1}{2}d(\cdot, R_0)^2$. The AIRM SPD manifold is Hadamard, and restriction to the isometric totally geodesic determinant-one fiber preserves the same constant Bhatia (2007); Lim (2004). A whitened tangent to the determinant-one hidden block has the form

$$Z = \begin{pmatrix} 0 & 0 \\ 0 & Z_{22} \end{pmatrix}, \quad Z_{22} = Z_{22}^\top, \quad \text{tr } Z_{22} = 0.$$

The whitened negative gradient is the tangent projection of $\mathcal{R}(\Sigma)$ in equation (29). Orthogonality to every trace-free Z_{22} is exactly $\mathcal{R}_{22}(\Sigma)^\circ = 0$. Strong convexity makes this condition necessary and sufficient.

Let $D = d_{\text{AI}}(\Sigma, \Sigma_\star)$, let $\xi = \text{Log}_\Sigma(\Sigma_\star)$, and let $V(\Sigma)$ be the restricted negative gradient. The first-order form of equation (81) gives

$$f_B(\Sigma) - f_B(\Sigma_\star) \leq \langle V(\Sigma), \xi \rangle - \frac{1}{2}D^2 \leq \|V(\Sigma)\| D - \frac{1}{2}D^2,$$

while the same inequality based at the minimizer, whose gradient vanishes, gives

$$f_B(\Sigma) - f_B(\Sigma_\star) \geq \frac{1}{2}D^2.$$

Adding the two bounds yields

$$D^2 \leq \|V(\Sigma)\| D,$$

and hence, also for $D = 0$,

$$D \leq \|V(\Sigma)\| = \|\mathcal{R}_{22}(\Sigma)^\circ\|_F.$$

Finally,

$$f_B(\Sigma) - f_B(\Sigma_\star) \leq \sup_{D \geq 0} \left(\|V(\Sigma)\| D - \frac{1}{2}D^2 \right) = \frac{1}{2} \|V(\Sigma)\|^2.$$

This proves the exact constants in equations (31) and (32). $\square$

B.1 Proof of active reduction

Let J be the orthogonal reflection that equals $+I$ on W and $-I$ on $W^\perp$. Since the columns of A and B lie in W ,

$$JA = A, \quad JB = B.$$

If $P \in \mathcal{F}_A(B)$, then $(JPJ)A = B$, and JPJ has the same determinant and distance from I as P . Uniqueness of the canonical controller yields

$$Js_A(B)J = s_A(B).$$

Hence the cross blocks between W and $W^\perp$ vanish: $s_A(B) = H_\star \oplus C_\star$.

For fixed feasible H , the inactive log-eigenvalues z_i satisfy

$$\sum_{i=1}^q z_i = -\log \det H.$$

Cauchy–Schwarz gives

$$\|\log C\|_F^2 = \sum_i z_i^2 \geq \frac{(\log \det H)^2}{q},$$

with equality exactly when all z_i are equal. Thus

$$C_\star = (\det H_\star)^{-1/q} I_q,$$

and substitution yields equation (34). Uniqueness follows from uniqueness of $s_A(B)$.

C Proofs for the Solver, Certificates, and Recovery Laws

This appendix proves the algorithmic and recovery results in sections 6 and 8. All gradients, exponential maps, and norms below use the affine-invariant metric.

C.1 Curvature and squared-distance comparison

Lemma 1 (AIRM curvature bound). *Every sectional curvature of $\mathbb{S}_{++}^d$, its determinant-one slice $\mathcal{P}_d$, and every action fiber $\mathcal{F}_A(B)$ lies in*

$$-\frac{1}{2} \leq \sec \leq 0. \quad (82)$$

Proof. Congruence invariance reduces the calculation to the identity, where tangent vectors are symmetric matrices and the metric is Frobenius. The AIRM curvature tensor is

$$R(X, Y)Z = -\frac{1}{4}[[X, Y], Z],$$

and hence

$$\sec(X, Y) = -\frac{1}{4} \frac{\| [X, Y] \|_F^2}{\|X\|_F^2 \|Y\|_F^2 - \langle X, Y \rangle_F^2} \leq 0.$$

To obtain the lower bound, replace Y by its component $\tilde{Y}$ Frobenius-orthogonal to X ; this leaves the commutator unchanged and turns the denominator into $\|X\|_F^2 \|\tilde{Y}\|_F^2$. Orthogonally diagonalize $X = \text{diag}(\lambda_i)$.

For symmetric $\tilde{Y} = (y_{ij})$,

$$\begin{aligned} \|[X, \tilde{Y}]\|_F^2 &= 2 \sum_{i < j} (\lambda_i - \lambda_j)^2 y_{ij}^2 \\ &\leq 4 \|X\|_F^2 \sum_{i < j} y_{ij}^2 \leq 2 \|X\|_F^2 \|\tilde{Y}\|_F^2. \end{aligned}$$

Substitution gives $\sec \geq -1/2$. The determinant-one slice and the action fibers are totally geodesic, so their intrinsic curvatures are restrictions of the ambient curvature tensor. $\square$

Lemma 2 (Squared-distance Hessian comparison). *Let $(\mathcal{M}, g)$ be a finite-dimensional Hadamard manifold with $-\kappa \leq \sec \leq 0$. For fixed z , let $h(x) = \frac{1}{2}d(z, x)^2$. Then*

$$g_x \preceq \text{Hess } h(x) \preceq \psi(\sqrt{\kappa}R)g_x \quad \text{whenever } d(z, x) \leq R, \quad (83)$$

where $\psi(s) = s \coth s$ and $\psi(0) = 1$. In particular, h is globally 1-strongly geodesically convex.

Proof. For $x \neq z$, put $r = d(z, x)$, let T be the terminal unit tangent of the geodesic from z to x , and decompose $v = aT + v_\perp$. The radial Hessian formula gives

$$\text{Hess } h(v, v) = a^2 + r \text{Hess } r(v_\perp, v_\perp).$$

Let J be the Jacobi field with $J(0) = 0$ and $J(r) = v_\perp$. Nonpositive curvature and Cauchy-Schwarz applied after parallel transport yield

$$\text{Hess } r(v_\perp, v_\perp) = I(J, J) \geq \int_0^r \|D_t J\|^2 dt \geq r^{-1} \|v_\perp\|^2.$$

For the reverse bound, compare J with

$$V(t) = \frac{\sinh(\sqrt{\kappa}t)}{\sinh(\sqrt{\kappa}r)} P_t v_\perp,$$

using $V(t) = (t/r)P_t v_\perp$ when $\kappa = 0$. The index lemma and the curvature lower bound give

$$I(J, J) \leq I(V, V) \leq \sqrt{\kappa} \coth(\sqrt{\kappa}r) \|v_\perp\|^2.$$

Therefore

$$\|v\|^2 \leq \text{Hess } h(v, v) \leq \psi(\sqrt{\kappa}r) \|v\|^2.$$

The formula extends continuously to $x = z$, where $\text{Hess } h = g_z$. Monotonicity of ψ proves the stated radius bound, and integration of the lower Hessian bound along geodesics proves strong convexity. $\square$

Proof of Theorem 6. Coercivity and closedness give a minimizer, and strong convexity makes it unique. Write $G_k = \text{grad } F(x_k)$, $g_k = \|G_k\|$, and

$$\gamma_k(t) = \text{Exp}_{x_k}(-tG_k), \quad 0 \leq t \leq L_0^{-1}.$$

If $g_k = 0$, strong convexity gives $x_k = x_*$. Otherwise $(F \circ \gamma_k)'(0) = -g_k^2 < 0$. If this update segment left the current sublevel, continuity would give a first return $t_* \in (0, L_0^{-1}]$. Before that return the Hessian is bounded by $L_0 g$, so the integral Taylor formula would give

$$\begin{aligned} F(\gamma_k(t_*)) &\leq F(x_k) - t_* g_k^2 + \frac{L_0 t_*^2}{2} g_k^2 \\ &\leq F(x_k) - \frac{t_*}{2} g_k^2 < F(x_k), \end{aligned}$$

contradicting the return. Hence the complete update stays in the current sublevel, and the same inequality at $t = L_0^{-1}$ gives

$$F(x_{k+1}) \leq F(x_k) - \frac{g_k^2}{2L_0}.$$

Let $\Delta(x) = F(x) - F(x_*)$, $d = d(x, x_*)$, and $g = \|\text{grad } F(x)\|$. Strong convexity at x and at x_* gives

$$\frac{\mu}{2} d^2 \leq \Delta(x) \leq g d - \frac{\mu}{2} d^2 \leq \frac{g^2}{2\mu}.$$

The first two expressions also imply $d \leq g/\mu$. Thus $g^2 \geq 2\mu\Delta$, and the descent inequality yields

$$\Delta_{k+1} \leq \left(1 - \frac{\mu}{L_0}\right) \Delta_k.$$

Iteration proves the value rate; the two preceding inequalities prove the distance and posterior bounds. Since $g_k^2/(2L_0) \leq \Delta_k$, they also give the gradient rate. Exact inversion of $q_{\mu,L}^k$ gives the stated complexity. If $L_0 = \mu$, the recurrence gives $\Delta_1 = 0$.

For $F(x) = \mu\|x\|^2/2$ on a Euclidean line, both posterior inequalities and the $L_0 = \mu$ one-step statement hold with equality. $\square$

For the fiber objective $F_B(P) = \frac{1}{2}d_{\text{AI}}(I, P)^2$, Lemmas 1 and 2 imply

$$\text{Hess } F_B \succeq g, \quad \text{Hess } F_B(P) \preceq \psi(D/\sqrt{2})g_P \quad \text{if } d_{\text{AI}}(I, P) \leq D. \quad (84)$$

The following action instance proves that the constant cannot be decreased uniformly over dimensions and admissible actions.

Proposition 1 (Action-family attainment of the radius majorant). *Let $d = 3$, $m = 1$, and $A = B = e_1$. The action fiber is*

$$\{\text{diag}(1, \Sigma) : \Sigma \in \mathcal{P}_2\},$$

its canonical controller is I_3 , and for every $D > 0$ there are a point at AIRM radius D and a unit fiber tangent v such that

$$\text{Hess } F_B(v, v) = \psi(D/\sqrt{2}). \quad (85)$$

Hence $\psi(D/\sqrt{2})$ is the sharp dimension-uniform radius-only Hessian majorant within the action family. Along a radial geodesic in the same fiber, both posterior inequalities in equations (49) and (50) hold with equality.

Proof. Here $M = 1$, $N = 0$, and $\rho = 1$, so the displayed fiber and canonical controller follow directly from equation (25). The determinant-one surface $\mathcal{P}_2$ is a two-dimensional space of constant sectional curvature $-1/2$, as equality holds in the commutator bound of Lemma 1. On a constant-curvature $-\kappa$ plane, the transverse eigenvalue of the Hessian of $\frac{1}{2}d(I, \cdot)^2$ at radius D is $D\sqrt{\kappa} \coth(D\sqrt{\kappa})$. Taking $\kappa = 1/2$ gives equation 85. $\square$

C.2 Residual gradient and exact global convergence

Proof of equation (41). On the ambient SPD manifold, for $\tilde{f}(Q) = \frac{1}{2}d_{\text{AI}}(R_0, Q)^2$,

$$-\text{grad } \tilde{f}(Q) = \text{Log}_Q(R_0).$$

Whitening at Q turns this tangent into $\log(Q^{-1/2}R_0Q^{-1/2})$. At $Q = Q_\Sigma$ it is precisely $\mathcal{R}(\Sigma)$. The whitened tangent space of the hidden determinant-one block consists of

$$\begin{pmatrix} 0 & 0 \\ 0 & Z \end{pmatrix}, \quad Z = Z^\top, \quad \text{tr } Z = 0.$$

Since whitening converts the AIRM inner product to the Frobenius inner product, orthogonal projection retains the 22 block and removes its trace. Thus the whitened reduced negative gradient is $W_B(\Sigma) = \mathcal{R}_{22}(\Sigma)^\circ$, which proves both identities in equation (41). $\square$

Proof of Theorem 7. Set

$$f_k = f_B(\Sigma_k), \quad g_k = \|W_B(\Sigma_k)\|_F, \quad D_k = d_{\text{AI}}(R_0, Q_{\Sigma_k}) = \sqrt{2f_k}.$$

Trace freeness of $W_B(\Sigma_k)$ preserves determinant one, while the exponential preserves positive definiteness. The action chart consequently keeps every controller on $\mathcal{F}_A(B)$.

Let $\gamma_k(t)$ be the negative-gradient geodesic, parameterized for $0 \leq t \leq L_0^{-1}$, and put $\varphi_k = f_B \circ \gamma_k$. If $g_k = 0$, the iterate is optimal. Otherwise $\varphi'_k(0) = -g_k^2 < 0$. If the update segment left the current objective sublevel, there would be a first return $t_* \in (0, L_0^{-1}]$. Before that return, $D(\gamma_k(t)) \leq D_k \leq D_0$, so equation 84 bounds the Hessian by $L_0 g$. The integral Taylor formula would then imply

$$\varphi_k(t_*) \leq \varphi_k(0) - t_* g_k^2 + \frac{L_0 t_*^2}{2} g_k^2 < \varphi_k(0),$$

a contradiction. Thus the complete update segment stays in the current sublevel. Applying the same Taylor bound at L_0^{-1} gives

$$f_{k+1} \leq f_k - \frac{1}{L_0} g_k^2 + \frac{1}{2L_0} g_k^2 = f_k - \frac{1}{2L_0} g_k^2.$$

Thus $D_{k+1} \leq D_k$, closing the induction and proving equation (45) globally.

Let $\Delta(\Sigma) = f_B(\Sigma) - f_B(\Sigma_*)$. Strong convexity and Cauchy–Schwarz give, with $d = d_{\text{AI}}(\Sigma, \Sigma_*)$ and $g = \|\text{grad } f_B(\Sigma)\|$,

$$\Delta(\Sigma) \leq gd - \frac{1}{2}d^2 \leq \frac{1}{2}g^2. \tag{86}$$

Hence $g^2 \geq 2\Delta$, and the descent inequality yields

$$\Delta_{k+1} \leq (1 - L_0^{-1})\Delta_k = q_0 \Delta_k.$$

Since $\Delta_0 \leq f_0 = D_0^2/2$, this proves equation (46). Strong convexity based at the minimizer gives $d^2 \leq 2\Delta$, proving equation (47). Combining $f_{k+1} \geq f_B(\Sigma_*)$ with the descent inequality gives $g_k^2 \leq 2L_0 \Delta_k$, proving equation (48).

The two strong-convexity inequalities also give $\Delta \geq d^2/2$. Together with the first bound in equation 86, this implies $d \leq g$, including the case $d = 0$, while the second bound in equation 86 gives $\Delta \leq g^2/2$. These are equations (49) and (50). Finally, for $D_0 > 0$, $q_0^k = e^{-k\chi_0}$. Solving the three rate bounds exactly for k gives equations (51) to (53). The case $D_0 = 0$ is immediate. $\square$

C.3 Active and inexact implementations

Proof of Corollary 2. Retain $A = UC$ from section 5. Choose V_a so that $E = [U, V_a]$ is an orthonormal basis of $\mathcal{W}_A$. Then

$$BC^{-1} = UM + V_a N_a, \quad T_a = N_a M^{-1}, \quad L_a = \begin{pmatrix} I & 0 \\ T_a & I \end{pmatrix}, \quad R_{0,a} = L_a^{-1} L_a^{-\top}.$$

Complete E by Z when $q > 0$. In the basis $[U, V_a, Z]$, the full action coordinate satisfies

$$N = \begin{pmatrix} N_a \\ 0 \end{pmatrix}, \quad L = L_a \oplus I_q, \quad R_0 = R_{0,a} \oplus I_q.$$

An active-compatible initializer has the form

$$\Sigma = \Sigma_a \oplus \beta I_q, \quad \det \Sigma_a \beta^q = 1.$$

The identity default corresponds to $\Sigma_a = I_s$ and $\beta = 1$. Then

$$\mathcal{Q}_a = \text{diag}(M, \rho \Sigma_a), \quad \mathcal{Q}_\Sigma = \mathcal{Q}_a \oplus (\rho \beta) I_q,$$

and block functional calculus gives

$$\mathcal{R}_a = \log(\mathcal{Q}_a^{-1/2} R_{0,a} \mathcal{Q}_a^{-1/2}), \quad \mathcal{R}(\Sigma) = \mathcal{R}_a \oplus r_c I_q, \quad r_c = -\log(\rho \beta).$$

The hidden trace-free projection subtracts the common mean

$$\tau = \frac{\text{tr } \mathcal{R}_{22}^a + q r_c}{s + q}.$$

Consequently,

$$\begin{aligned} W_B(\Sigma) &= W_a \oplus w_c I_q, & W_a &= \mathcal{R}_{22}^a - \tau I_s, & w_c &= r_c - \tau, \\ \|W_B(\Sigma)\|_F^2 &= \|W_a\|_F^2 + q w_c^2. \end{aligned} \tag{87}$$

The full exponential update now splits exactly into

$$\Sigma_a^+ = \Sigma_a^{1/2} e^{W_a/L_0} \Sigma_a^{1/2}, \quad \beta^+ = \beta e^{w_c/L_0}. \tag{88}$$

Since $\text{tr } W_a + q w_c = 0$, this update preserves $\det \Sigma_a \beta^q = 1$. The action-bundle map then preserves positive definiteness, determinant one, and $PA = B$. This proves equation 87 and equation 88; the stopping certificate follows from equation (49).

A rank-revealing thin QR of the $d \times 2m$ input costs $O(dm^2)$. All remaining preprocessing acts on matrices of order at most r , costing $O(r^3)$. Each iteration uses one $r \times r$ logarithm and inverse square root and one $s \times s$ exponential, hence $O(r^3)$ arithmetic and $O(r^2)$ working memory. Combining this cost with equation (53) gives equation (55). The representation

$$Px = cx + E(H - cI_r)E^\top x$$

costs $O(dr + r^2)$ per application. If $q = 0$, every complement term is absent and the same construction is the full solver. If $s = 0$, the determinant condition forces $\beta = 1$, leaving no variable. $\square$

Proof of Corollary 3. Put $W_k = W_B(\Sigma_k)$, $g_k = \|W_k\|_F$, and $\widetilde{W}_k = W_k + E_k$. First, equation (56) and $b_k \geq \|E_k\|_F$ give

$$\|E_k\|_F \leq \frac{\delta}{1 + \delta} (\|W_k\|_F + \|E_k\|_F),$$

so rearrangement proves equation (57). Consequently,

$$(1 - \delta)g_k \leq \|\widetilde{W}_k\|_F \leq (1 + \delta)g_k, \quad \langle -W_k, \widetilde{W}_k \rangle_F \leq -(1 - \delta)g_k^2.$$

If $g_k = 0$, the relative error forces a zero update. Otherwise consider the inexact update geodesic. If it left the current objective sublevel, there would be a first return $t_\star \in (0, \eta]$. Before that return, the identity radius is at most $D_k \leq D_0 \leq \widehat{D}_0$, so the Hessian is bounded by $\widehat{L}_0 g$. The integral Taylor bound would give

$$f_B(\gamma_k(t_\star)) \leq f_k - t_\star(1 - \delta)g_k^2 + \frac{\widehat{L}_0 t_\star^2}{2}(1 + \delta)^2 g_k^2 < f_k,$$

contradicting the return. Hence the whole segment remains in the current sublevel. The descent lemma at $t = \eta$ implies

$$f_{k+1} \leq f_k - \eta(1 - \delta)g_k^2 + \frac{\widehat{L}_0 \eta^2}{2}(1 + \delta)^2 g_k^2 = f_k - \frac{c_\delta}{2\widehat{L}_0} g_k^2.$$

Hence $D_{k+1} \leq D_k$. Applying the global PL inequality from the exact proof gives

$$\Delta_{k+1} \leq \widehat{q}_\delta \Delta_k.$$

Iteration and $\Delta_0 \leq D_0^2/2 \leq \widehat{D}_0^2/2$ prove equation (59); strong convexity gives equation (60). The descent bound and $f_{k+1} \geq f_B(\Sigma_\star)$ imply

$$g_k^2 \leq \frac{2\widehat{L}_0}{c_\delta} \Delta_k.$$

Together with $\|\widetilde{W}_k\|_F \leq (1 + \delta)g_k$, the value rate, and $\sqrt{c_\delta} = (1 - \delta)/(1 + \delta)$, this proves equation (61).

Finally, the exact posterior certificates and $g_k \leq \|\widetilde{W}_k\|_F / (1 - \delta)$ give equation (62). $\square$

C.4 Nested recovery

Proof of Theorem 8. For every k ,

$$A_k^\top B_k = c_H(S^{(k)})^\top H S^{(k)} \succ 0,$$

so $\mathcal{C}_k$ is nonempty, closed, and totally geodesic by Theorem 5. Also $P_\star A_k = c_H H^{-1} H S^{(k)} = B_k$, so $P_\star \in \mathcal{C}_k$. Nested column spaces imply $\mathcal{C}_{k+1} \subset \mathcal{C}_k$. Metric projection onto a nonempty closed geodesically convex subset of a Hadamard manifold exists uniquely, proving that every update is well defined and retains every earlier action.

For completeness, if $y = \text{Proj}_C(x)$ and $z \in C$, first variation along the geodesic from y to z gives $\langle \text{Log}_y(x), \text{Log}_y(z) \rangle \leq 0$. Since the exponential map is distance expanding,

$$\begin{aligned} d(x, z)^2 &\geq \|\text{Log}_y(x) - \text{Log}_y(z)\|^2 \\ &\geq d(x, y)^2 + d(y, z)^2. \end{aligned}$$

Apply this with $x = P_{k-1}$, $y = P_k$, $z = P_\star$, and $C = \mathcal{C}_k$ to obtain equation (66). Summation telescopes to equation (67); Cauchy–Schwarz then gives equation (68).

If $r_K = d - 1$, the bundle fiber is diffeomorphic to $\mathcal{P}_{d-r_K} = \mathcal{P}_1 = \{1\}$, so $\mathcal{C}_K = \{P_\star\}$. If $r \leq d - 2$, then $n = d - r \geq 2$ and the fiber is diffeomorphic to $\mathcal{P}_n$, whose dimension $n(n + 1)/2 - 1$ is positive; it therefore contains infinitely many completions. This proves the sharp threshold and the stated block count.

Finally, for the k -th projection set

$$\widetilde{P} = P_{k-1}^{-1/2} P P_{k-1}^{-1/2}, \quad \widetilde{A}_k = P_{k-1}^{1/2} A_k, \quad \widetilde{B}_k = P_{k-1}^{-1/2} B_k.$$

Then $P A_k = B_k$ is equivalent to $\widetilde{P} \widetilde{A}_k = \widetilde{B}_k$, determinant one is preserved, and congruence invariance turns the objective into $d_{\text{AI}}(I, \widetilde{P})$. Thus every outer update is an identity-reference action problem. The exact projection is the limit covered by Theorem 7; Corollary 3 certifies finite inner approximations but is not used in the exact Pythagorean identities. $\square$

Proof of Theorem 9. Fix a round and abbreviate

$$x = \tilde{P}_{k-1}, \quad y = Q_k, \quad \tilde{y} = \tilde{P}_k, \quad z = P_\star,$$

and let $a = d_{\text{AI}}(x, y)$, $b = d_{\text{AI}}(y, z)$, $e = d_{\text{AI}}(y, \tilde{y})$, and $D = d_{\text{AI}}(x, z)$. Because y is the projection of x onto $\mathcal{C}_k$ and $z \in \mathcal{C}_k$, the CAT(0) projection inequality gives

$$a^2 + b^2 \leq D^2.$$

The triangle inequality and $e \leq \varepsilon_k$ give

$$\begin{aligned} d_{\text{AI}}(x, \tilde{y})^2 + d_{\text{AI}}(\tilde{y}, z)^2 &\leq (a + e)^2 + (b + e)^2 \\ &\leq D^2 + 2e(a + b) + 2e^2 \\ &\leq D^2 + 2\sqrt{2}eD + 2e^2, \end{aligned}$$

where the last line uses $a + b \leq \sqrt{2(a^2 + b^2)} \leq \sqrt{2}D$. This proves equation (72). Also

$$D_k \leq b + e \leq D_{k-1} + \varepsilon_k,$$

and induction proves equation (73). Rearranging equation (72) and summing proves equation (74). Nested feasibility preserves the earlier actions. The stated observable bounds are the posterior certificates after the congruence in the preceding proof. Finally, $r_K = d - 1$ makes the final fiber a singleton by Theorem 8. $\square$

D Closest-Work Comparison

The solver theorem combines standard Hadamard first-order principles with an action-specific geometric construction. Table 1 separates those responsibilities. It compares theorem outputs, not algorithms with different feasible models.

The residual-oracle corollary assumes rigorous direction-error and radius majorants. It propagates those certificates through an exact exponential update; unlike a certified numerical linear-algebra package, it does not enclose update, factorization, basis, storage, or determinant errors and does not establish bit complexity.

Work	Established output	Action-specific distinction
Hadamard first-order methods Zhang & Sra (2016)	Global complexity for smooth or nonsmooth geodesically convex objectives, including strongly convex and curvature-dependent regimes	No parameterization of $PA = B$, logarithmic action residual, or active controller realization
Hadamard convex analysis Bačak (2014)	Projection, proximal, and convex-analytic foundations on closed geodesically convex sets	Supplies abstract projection principles; does not construct or solve the action fiber
Tian Tian (2013)	Equalities, inequalities, existence conditions, and parameterizations for Hermitian and definite solutions of matrix equations including $AX = B$	Establishes the algebraic feasibility layer; does not select the determinant-one AIRM nearest solution or derive a residual iteration
Lim Lim (2004)	Best approximation in Riemannian geodesic SPD submanifolds, with explicit formulas for selected involutive block models	Supplies nearest-point precedents; does not identify all $PA = B$ fibers as one analytic bundle or give the action normal equation and active solver
Tumpach–Larotonda Tumpach & Larotonda (2024)	General criteria and structure for totally geodesic submanifolds of the SPD manifold	Supplies general structural tools; does not construct the action-indexed fiber family, its analytic section, or its posterior certificates
Conic SPD optimization Sra & Hosseini (2015)	Algorithms for geodesically convex conic optimization over SPD matrices	Does not specialize the $PA = B$ fiber into the residual, active reduction, or finite identification law
Least-change and multi-secant updates Dennis & Schnabel (1979); Schnabel (1983); Fletcher (1991); Gao & Goldfarb (2018); Boutet et al. (2022); Kanamori & Ohara (2013)	Update formulas and convergence properties for secant equations under method-specific norms or divergences	Do not parameterize the complete AIRM action fiber or prove the normalized rank- $d - 1$ identification threshold
This paper	For $PA = B$, $P \succ 0$, $\det P = 1$: global action bundle, total geodesy, analytic canonical section, exact residual, global rate, posterior certificates, and active total complexity	Specializes the general principles into an exact action solver with posterior certificates and conditional residual-error propagation

Table 1: Theorem-level comparison with the closest geometric optimization and multi-secant baselines.